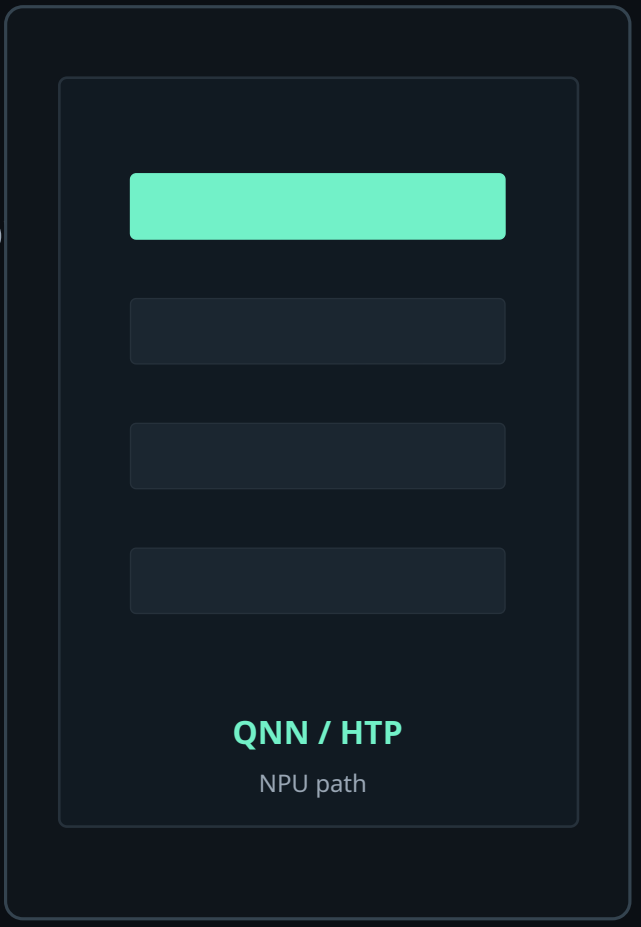
SNAPDRAGON AI LAB BUILD & PRESENT

SnapShield AI

On-device privacy redaction for Snapdragon-powered Windo PCs

Protect sensitive visuals before they leave the device.

Detect locally -> redact -> inspect -> export



Vinay Kumar

01 / PROBLEM

Sensitive pixels are shared too easily

Users need a quick privacy layer before an image is uploaded, emailed, or posted.

- Screenshots and photos can expose faces and other personal visual information.
- Cloud processing introduces an external dependency for a task that can be performed locally.
- Manual blur/redaction is slow and easy to miss on repeated workflows.
- The product should make privacy protection simple enough to use before every share.

LOCAL
source

FAST
workflow

CLEAR
feedback

Design goal

Make privacy redaction a one-click local operation with visible evidence of what happened.

SnapShield turns privacy into a local pipeline

A single interface handles detection, redaction, inspection, and export.

1

UPLOAD

Choose an image

2

DETECT

Faces + selected objects

3

REDACT

Blur sensitive regions

4

VERIFY

Side-by-side preview

5

EXPORT

Download safe copy

Privacy policies People + faces • People + devices • Maximum privacy

One app, two execution paths

Snapdragon hardware gets a first-class QNN/HTP path while the product stays portable during development.



QNN-first provider selection

The runtime asks ONNX Runtime whether QNN is available. On Snapdragon, QNN can target the HTP/NPU backend; otherwise the CPU execution provider keeps development portable.

Designed for Snapdragon NPU execution

The prototype uses an ONNX + QNN architecture instead of hard-wiring the experience to one host runtime.

Implementation choices

- ONNX format for portable model deployment.
- ONNX Runtime QNN Execution Provider selected first when available.
- HTP backend used as the Snapdragon NPU-oriented target.
- CPU fallback for development and compatibility.
- Runtime provider and latency surfaced in the interface.

AI Hub reference profile

YOLOv5 • Snapdragon X Elite CRD • Windows 11

14.8 ms estimated inference

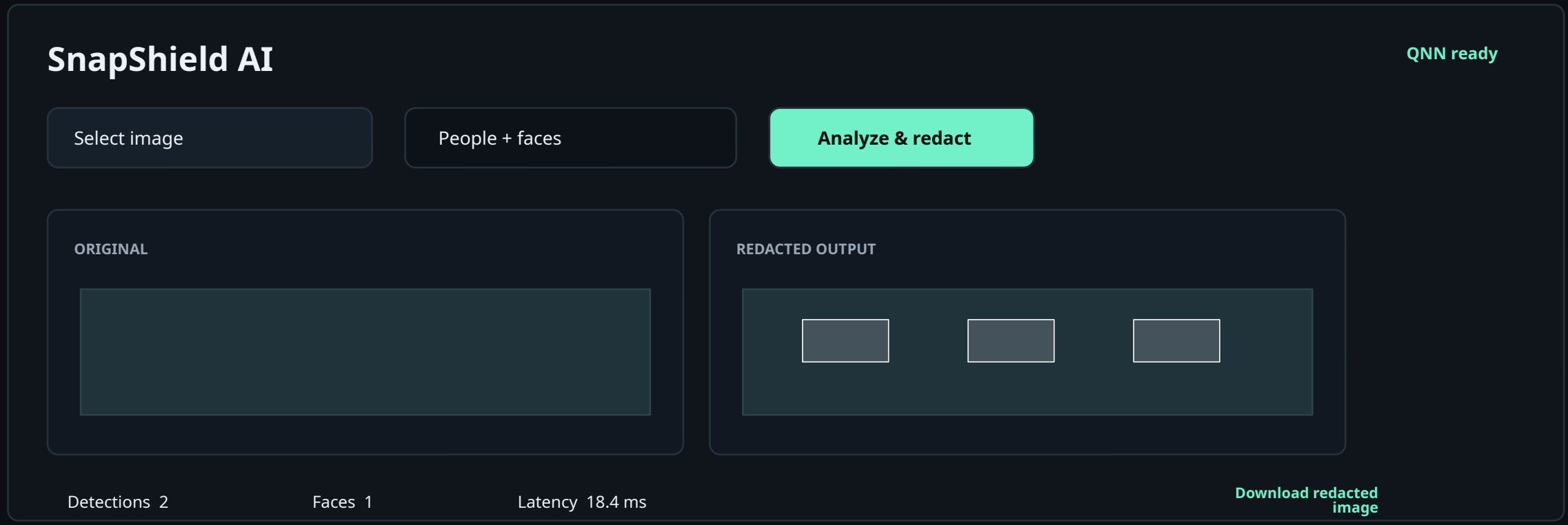
47 MB peak memory • NPU compute

Claim boundary

These numbers are Qualcomm AI Hub reference measurements for the published profile, not measurements claimed for this prototype.

A focused MVP built for a live demo

The experience is intentionally small: one image, one policy, one redacted result, and visible runtime telemetry.



UI mockup for pitch presentation - final device benchmark should be captured on the target Snapdragon PC.

From prototype to Windows product

The MVP establishes the architecture; the next iteration closes the hardware-validation and packaging loop.

NOW

- Local image redaction
- QNN-first runtime path
- Runtime telemetry
- Side-by-side verification

NEXT

- Benchmark on physical Snapdragon PC
- Add QR / OCR document redaction
- Package as Windows executable
- Add audit log / batch mode

VALUE

- Privacy before sharing
- No cloud upload in core flow
- Hardware-aware deployment
- Simple, inspectable UX

Core product promise: make sensitive pixels safer before they are shared.

REFERENCES

Technical references used for the pitch

Key sources for the Snapdragon deployment story and current challenge requirements.

Qualcomm Snapdragon AI Lab	https://www.qualcomm.com/snapdragon/ai-lab
Qualcomm AI Hub YOLOv5 profile	https://aihub.qualcomm.com/jobs/jpxl31x1p
ONNX Runtime QNN Execution Provider	https://onnxruntime.ai/docs/execution-providers/QNN-ExecutionProvider.html
Challenge listing	https://unstop.com/competitions/1748893
Ultralytics YOLOv5 / ONNX documentation	https://github.com/ultralytics/yolov5

Note: AI Hub performance figures are cited as reference data. No device-specific benchmark is represented as measured by SnapShield AI until validated on the target Snapdragon hardware.